Beyond Bigger: Quantum, AI, and the Future of Research Computing

Summary of the ERN Quantum Webinar by **Dr. Travis L. Scholten**, Technical Lead, Public Sector, IBM Quantum (IBM T. J. Watson Research Center) — 16 October 2026, 12:00–1:00 pm ET. Hosted by the Northeast Big Data Innovation Hub and the Ecosystem for Research Networking (ERN). Prepared by David Grunwald from the slides and live notes.

KEY IDEA

The thesis in one line: the next leap in research computing is not just *bigger* machines. It is **CPU + GPU + QPU working together** (quantum-centric supercomputing), and the hard part is shifting from hardware to **software, workflows and orchestration**. Quantum is a co-processor, not a replacement for classical computing.

1 Three Shifts Reshaping Computing

Shift	Meaning	Talk points
Computing is more essential	Why leaders care	Compute capacity is becoming foundational infrastructure. HPC centers, hyperscale data centers and "AI factories" are pieces of one foundation. *AI made computation visible; quantum makes it even more strategic.*
Computing is more integrated	AI, HPC and quantum together	CPU + GPU + QPU = **quantum-centric supercomputing**. Quantum augments classical computing and extends the frontier alongside GPUs.
Discovery is more collaborative	Spanning machines, methods and institutions	The frontier is bigger than any one institution (RIKEN, Cleveland Clinic, Lockheed Martin, ORNL, RPI, Basque Quantum, National Quantum Algorithm Center).

- **Research institutions are becoming builders** of computational infrastructure, not just consumers. Knowing how to access and deploy it is a strategic edge, an "innovation flywheel."
- **National priority:** U.S. Executive Order 14413, "Ushering in the Next Frontier of Quantum Innovation" (22 June 2026), sets a whole-of-government approach. Also cited: the **DOE Genesis Mission** (a new substrate for scientific discovery) and **DARPA QBI** (Quantum Benchmarking Initiative, validating scalable quantum computing).

2 Where Quantum Stands Today

Year	IBM milestone	Detail
2016	Quantum is online	IBM puts quantum devices on the cloud
2017	Qiskit SDK launched	Open-source quantum software
2023	Quantum utility	5K operations on 100+ qubits
2026	Quantum advantage (target)	With clear separation and verifiability
2029	First fault-tolerant computer (target)	100M operations on 200 logical qubits (Starling)

- **Access is no longer the bottleneck.** IBM runs 2 quantum facilities (Poughkeepsie, USA: 6 systems; Ehningen, Germany: 3 systems, all 100+ qubits) plus 9 (+1) on-site systems: Univ. of Tokyo, Cleveland Clinic, PINQ^2 (Canada), RPI, Yonsei, RIKEN (Kobe), IBM-Euskadi (San Sebastián), with the National Quantum Algorithm Center (Chicago, 3Q 2026) and Quantum Valley Tech Park (India, 2H 2026) projected.

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- **Today's machines are noisy**, not yet fault-tolerant. The emphasis is on **validating quantum results against classical ones**, so quantum output is checkable rather than a "credence good" taken on faith.
- **Evidence of value:** empirical usefulness tests (HSBC, Cleveland Clinic, Moderna, Vanguard) and scientifically rigorous advantage claims (ORNL, RIKEN, Los Alamos, BasQ, Flatiron Institute, BlueQubit). Slide links are in Section 7.

Not every research problem needs quantum

Scholten's screening pipeline: **Research Problem → Computational Bottleneck → Problem Class → Classical Baseline → Quantum Investigation.** Do the prework to decide whether quantum is useful before using it.

Strong candidates	Weak candidates
Approximate solutions are acceptable	Poorly defined classical baselines
Fits existing HPC workflows	Tiny problem scales
Meaningful operational impact	No operational impact

Problem classes mentioned in the talk: Hamiltonian/materials simulation, chemistry (DFT, diagonalization), optimization (e.g., traveling salesman, where solutions blow up), differential equations (e.g., Navier–Stokes; "Nature does!"), and quantum-enhanced ML (classifiers, mixture-of-experts models, generative modeling). Quantum is *one part* of the solution, aimed at pieces not amenable to classical methods.

3 Scaling Is Systemic: The Hardware Roadmap

Progress now goes beyond processor size: it is the shift from **science to engineering**. Build a logical-level building block, then build and connect several.

System (year)	Key features
Loon (2025)	2 logical qubits; 6-way connectivity; c-coupler demo; automated design
Kookaburra (2026)	"Long" c-couplers; real-time decoding; LPU (+~100 qubits) for logical operations; logical memory
Cockatoo (2027)	Two blocks of gross code + LPU; module-to-module logical communication over l-couplers
Starling (2028–2029)	Multiple gross-code blocks + LPUs; universal computation with a **magic state factory** (magic state distillation). 200 logical qubits, 100M gates
Blue Jay (beyond)	2,000 logical qubits, 1 billion gates
System Two (2025)	Supports 1,000+ physical qubits, 15,000+ gates (4× in the IBM Quantum Data Center)

> **NOTE ON THE SLIDES**
> The slides date Starling as both 2028 (milestones slide) and 2029 (data-center and maturity slides). 2029 is IBM's stated fault-tolerant delivery date.

- **Modular cryogenic architecture:** cryo cells are built, tested, shipped and assembled; cooling, wiring, electronics and quantum devices evolve independently. Early systems: two modules, ~1,000 qubits, designed to scale to dozens of modules and hundreds of thousands of physical qubits.
- **Manufacturing:** moving to **300 mm wafer processing**: qLDPC codes need >~10× more chip complexity; a 24/7 300 mm fab line gives a 2–3× speedup per wafer; >11,000 quantum wafers have run through Albany Nanotech. Wafer images compare the Heron and Loon chips.

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- **Anderon:** a new pure-play quantum foundry from IBM, with U.S. Department of Commerce support, to build world quantum manufacturing capability (fab more chips, quicker, with good yields).
- **Power:** Heron/Nighthawk systems use about 50–100 kW (Nighthawk ~55 kW). Targets are 1.5 MW for Starling and 6.3 MW for Blue Jay, versus hundreds of MW to GWs for modern data centers. Quantum does not have the same facility needs as AI data centers, so computing facilities can bring it in more easily, and it may *reduce* classical resource use for some steps.

4 Quantum-Centric Supercomputing (QCSC)

Phase	Architecture	Features
~2026–2028	Quantum as a **co-processor** to HPC	Specialized compute offload engines; co-location with existing HPC; post-selected QEC
~2028–2031	**Heterogeneous** quantum + HPC	Tight coupling; advanced middleware for hybrid algorithms; unified scheduling and resource management; interoperable fabric; near-time conditional QEC
~2031–2034	**Tightly integrated** quantum + HPC	Co-designed systems; unified programming model; high-performance quantum–classical fabric; multi-tenant execution; real-time QEC

The quantum–classical workflow

- **Pre-process and optimize** for quantum hardware (CPUs/GPUs).
- **Execute** on quantum hardware (QPU), with an **inner code** plus **error correction** in real/near time on FPGAs and ASICs.
- **Post-process** results, including **error mitigation** (CPUs/GPUs).
- **Orchestrate** the loop on classical hardware, then feed results back.
- **Workflows, not single programs**, become the core unit of scientific computing: Research Question → orchestrated compute loop → Scientific Insight.
- **Orchestration is the new software challenge:** managing an adaptive workflow over hardware that is data-bound or compute-bound and runs at different clock speeds. AI may help ("AI for quantum"). Quantum error correction will demand even closer coordination.
- **Software stack (top to bottom):** Qiskit Functions → algorithms and applications (Qiskit addons and ecosystem) → workflow management + frameworks (Qiskit SDK, Qiskit C API) → resource management (**Slurm quantum plugin**) → IBM Quantum Compute Service / QPU alongside CPU, GPU and AIU.
- **QRMI (Quantum Resource Management Interface):** an open, thin, vendor-agnostic layer (Rust core, Python and C bindings) to acquire, run tasks on and meter on-prem or cloud quantum systems. It has HPC plugins for **Slurm** and **Prefect** and supports IBM Quantum, Pasqal and AQT. Contributors include IBM, RPI, Pasqal, STFC Hartree, Cleveland Clinic, ORNL, Jülich, Poznan (PSNC) and ParTec.
- **Example integration:** RPI's IBM Quantum System One (127-qubit Eagle) is linked to the AiMOS supercomputer 10 miles away over 10 Gb/s Ethernet. Several on-site deployments (e.g., RIKEN) now run on-campus quantum alongside HPC and control access locally.

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5 Science Case Studies

Who	What	Result
BasQ + IBM	Hamiltonian simulation of quarks and strong-force confining strings	144 qubits, 7,862 two-qubit gates; real-time string dynamics
RIKEN + IBM	Chemistry beyond exact diagonalization; closed-loop electronic structure on QPU + classical supercomputer	Introduced **SQD**. Fe_4S_4 on 72 qubits: ~2 hours (QCSC) vs. ~3M years (VQE) vs. 4.53M qubits / 13 days (fault-tolerant phase estimation). Extrapolated energies ~0.1 Hartree from classical SOTA DMRG
ORNL + Cleveland Clinic + IBM	Molten salts and tritium for fusion-reactor blankets (DOE Genesis Mission)	9 molecular configurations; first such computations on quantum hardware
ORNL + Purdue + collaborators	Neutron scattering vs. quantum simulation of **$KCuF_3$** magnetism	Close agreement between experiment (6 K) and IBM-computed spectra; a hybrid route to faster materials discovery
Cleveland Clinic + RIKEN + IBM	Drug-relevant molecules using 2 QPUs + 2 supercomputers	Up to **12,635 atoms** (trypsin): the largest known biologically meaningful simulation on quantum hardware

Cleveland Clinic's path to drug discovery (SQD progression)

- First SQD on organic molecules (non-covalent interactions) → SQD integrated with **DMET** for large molecules → SQD extended to the liquid state (solute–solvent) → open-source Qiskit Function template with an implicit solvent model → fragmentation methods to cut qubit counts → Trp-cage mini-protein (303 atoms) → protein–ligand complexes (11K/12K atoms).
- Scale growth: methane dimer (10 atoms, Oct 2024) → cyclohexane (36) → Trp-cage (303, Dec 2025) → T4-lysozyme (11,608, Feb 2026) → trypsin (12,635, Apr 2026).

CONNECTION TO THE TUTORIAL

The ORNL/Purdue $KCuF_3$ slide is the paper behind the Quantum 16-qubit Application tutorial: Lee, Kumaran, et al., "Benchmarking quantum simulation with neutron-scattering experiments" (arXiv:2603.15608). The paper uses up to 50 qubits on IBM hardware to compute the dynamical structure factor S(q, ω) of a 1D Heisenberg chain and benchmarks it against inelastic neutron scattering. Co-author Travis Humble (ORNL) leads the Quantum Science Center. Quote from the slide (Arnab Banerjee, Purdue): "Using a quantum computer to compare with this data has been a goal for many years and we have now shown it is possible."

6 Communities and Collaboration

- **Communities enable scale:** 50+ active industry prototypes and Road to Utility projects; 437 Road to Practitioner students (RPI, FreeYa Mind, Yonsei, Univ. of Tokyo, BasQ); 20K+ participants in the Qiskit Summer School, Challenge and Fall Fest.
- **Technical working groups:** healthcare and life sciences, sustainability, optimization, materials science and high-energy physics.
- **Discussion themes:** what is the role of fundamental science in an engineering environment? How can academia meaningfully inform industry (e.g., Fermilab's SQMS center), and how do the two avoid getting in each other's way?

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- **Next in the series:** "From Quantum Discovery to Quantum Infrastructure," Michael Eggleston, PhD (Head, Nokia Bell Labs Quantum Program), 11 December 2026, 12:00–1:00 pm ET.

7 Links and References Presented

Links are given as shown on the slides. Bare arXiv numbers are linked to arxiv.org.

Topic (slide)	Reference
QRMI open standard	github.com/qiskit-community/qrmi
Qiskit Function template (Cleveland Clinic)	ibm.biz/ccf-function-template
$KCuF_3$ neutron scattering (ORNL/Purdue)	arXiv:2603.15608
Molten salts / fusion (ORNL, Cleveland Clinic)	arXiv:2606.30402 ibm.com/quantum/blog/molten-salts-fusion-quantum
Largest biomolecule simulation (Cleveland Clinic/RIKEN)	arXiv:2605.01138 ibm.com/quantum/blog/cleveland-clinic-riken-chemistry
BasQ confining strings	arXiv:2507.08088 ibm.com/quantum/blog/basque-quantum-leadership
RIKEN: Chemistry beyond exact solutions (SQD)	Science Advances 11(25), 18 Jun 2025, DOI 10.1126/sciadv.adu9991; also arXiv:2405.05068
RIKEN: closed-loop electronic structure	arXiv:2511.00224
Cleveland Clinic: supramolecular interactions	arXiv:2410.09209; Communications Physics 8, 396 (2025)
Lockheed Martin: methylene singlet/triplet	arXiv:2411.04827
ORNL: Sample-based Krylov diagonalization	arXiv:2501.09702
QCSC for materials science (perspective)	FGCS 160, 666–710 (2024); arXiv:2312.09733
Cleveland Clinic SQD series	J. Chem. Theory Comput. 21, 14 (2025); J. Phys. Chem. B 129, 23 (2025); arXiv:2506.20825; arXiv:2512.17130
Usefulness tests: HSBC / Moderna / Vanguard	arXiv:2509.17715; arXiv:2405.20328; arXiv:2508.13557
Advantage proofs: LANL	Nature Computational Science (2025)
Quantum-enabled cell-centric therapeutics	arXiv:2307.05734
Quantum Optimization Benchmarking Library	arXiv:2504.03832
Quantum computing for high-energy physics (QC4HEP)	PRX Quantum 5, 037001 (2024); arXiv:2307.03236
Host community (NE Big Data Hub / ERN)	Slack: bit.ly/nsdc-slack-channel, bit.ly/cic-slack-channel; YouTube: bit.ly/NEastYT; Instagram @nebigdatahub

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Topic (slide)	Reference
Related tutorial and IBM guides	programmar7.wordpress.com — Quantum 16-qubit Application IBM tutorial: simulate neutron scattering

NOTE ON THE SLIDES

Two slide typos are corrected here: "arXiv:2405:05068" → 2405.05068 and "arXiv:2410:09209" → 2410.09209. The policy, government and company program names are recorded as presented and were not independently checked.

8 Concepts and Definitions

Term	Definition
Quantum-centric supercomputing (QCSC)	IBM's model of computing that combines CPUs, GPUs and QPUs in one orchestrated workflow, with quantum as one accelerator among several.
QPU	Quantum processing unit: the quantum chip plus its control system, used like a GPU-style accelerator.
AIU	IBM's AI accelerator unit, listed beside CPU/GPU as a classical resource.
Heterogeneous computing	Systems that mix different processor types (CPU, GPU, QPU, FPGA, ASIC) and assign each task to the best one.
Quantum co-processor	A QPU used for specific offloaded subproblems while classical HPC runs the rest of the workflow.
Orchestration	Software that schedules, moves data between and coordinates steps across heterogeneous resources; called "the new software challenge."
Workflow-centric computing	Treating multi-step pipelines, not single programs, as the core unit of scientific computing.
AI factories	Large AI-dedicated data centers, seen as part of the same foundational compute infrastructure as HPC centers.
Innovation flywheel	Access to advanced compute → deployment knowledge → results → more investment and access: a self-reinforcing strategic edge.
Quantum utility	Quantum computers producing reliable results at a scale beyond brute-force classical simulation (IBM, 2023: 5K operations, 100+ qubits).
Quantum advantage	A quantum computation that beats the best classical methods, with clear separation and verifiable results.
Credence good	Something whose quality the buyer can't verify. The talk stressed that validation against classical results keeps quantum output from being one.
Classical baseline	The best classical method for a problem; needed to judge whether quantum adds value.
Noisy / pre-fault-tolerant	Today's hardware: errors accumulate, so results need mitigation and validation.
Fault-tolerant quantum computer	A computer whose error correction keeps logical errors low enough to run very long algorithms (Starling target: 100M gates).
Physical vs. logical qubit	A physical qubit is one hardware qubit. A logical qubit is an error-protected qubit encoded across many physical qubits.
QEC (quantum error correction)	Detecting and correcting errors during computation. Roadmap: post-selected → near-time conditional → real-time.

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Term	Definition
Error mitigation	Post-processing that reduces the effect of noise on results without full error correction.
Real-time decoding	Classical hardware (FPGAs/ASICs) that interprets error syndromes fast enough to correct errors while the computation runs.
Inner code	The error-correcting code running directly on the QPU in the execution step.
qLDPC / gross code	Quantum low-density parity-check codes; the gross code is IBM's qLDPC code, which needs far fewer physical qubits per logical qubit than surface codes.
LPU	Logical processing unit: the block that performs operations on logical qubits stored in logical memory.
c-coupler / l-coupler	Long-range on-chip couplers (c) enabling qLDPC connectivity, and inter-module links (l) enabling logical communication between modules.
Magic state factory	A dedicated region that prepares high-fidelity "magic states" (via distillation), enabling universal fault-tolerant computation.
Modular cryogenic architecture	Cryostats built as shippable cells that are assembled into larger systems, with subsystems upgradable independently.
300 mm wafer processing	Industry-standard semiconductor wafer size; adopted to make more and more complex quantum chips faster with good yield.
Anderon	IBM's new pure-play quantum foundry, supported by the U.S. Department of Commerce.
Heron / Nighthawk / Loon / Kookaburra / Cockatoo / Starling / Blue Jay	IBM processor and system generations, from current processors (Heron, Nighthawk) through fault-tolerance demonstrations to large fault-tolerant machines.
IBM Quantum System Two	IBM's modular system housing for multiple processors (1,000+ physical qubits).
Slurm / Slurm quantum plugin	The common HPC job scheduler, and the plugin that lets it schedule quantum jobs alongside classical ones.
QRMI	Quantum Resource Management Interface: an open, vendor-agnostic layer to acquire, run tasks on and meter quantum resources from HPC.
Prefect	A Python workflow-orchestration framework with a QRMI plugin.
Qiskit SDK / C API / Functions / addons	IBM's open-source quantum toolkit, its C interface for HPC codes, managed application services, and research add-on modules.
Multi-tenant execution	Several users or jobs sharing a quantum system at once.
Co-location	Installing a QPU on-site next to HPC systems to cut latency and give local access control.
SQD	Sample-based quantum diagonalization: the QPU samples important electron configurations, then a classical supercomputer diagonalizes the Hamiltonian in that subspace.
SKQD	Sample-based Krylov quantum diagonalization: an SQD variant using Krylov-space states (ORNL).
VQE	Variational quantum eigensolver: an earlier near-term approach; shown as impractically slow (~3M years) for Fe_4S_4.
Phase estimation	The fault-tolerant algorithm for energies; estimated at 4.53M qubits and 13 days for Fe_4S_4.
DMET	Density matrix embedding theory: splits a large molecule into fragments, solving the hard fragment quantum-mechanically.
Fragmentation methods	Breaking a molecule into pieces so fewer qubits are needed without losing accuracy.

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Term	Definition
DFT	Density functional theory: a standard classical chemistry method that quantum may complement.
DMRG	Density matrix renormalization group: a leading classical tensor-network method, used as the state-of-the-art benchmark.
Hartree	An atomic unit of energy (~27.2 eV); RIKEN's extrapolated energies were ~0.1 Hartree from DMRG.
Hamiltonian simulation	Simulating a physical system's time evolution under its energy operator; a natural fit for quantum computers.
Dynamical structure factor S(q, ω)	The momentum- and energy-resolved spectrum that neutron scattering measures; computed on quantum hardware for $KCuF_3$.
Inelastic neutron scattering	An experiment in which neutrons exchange energy and momentum with a material's spins, revealing its excitations.
EO 14413	U.S. Executive Order (22 June 2026) establishing a whole-of-government approach to quantum.
DOE Genesis Mission	A U.S. Department of Energy program for AI- and compute-driven scientific discovery; it funded the molten-salt fusion work.
DARPA QBI	Quantum Benchmarking Initiative: DARPA's program to validate whether scalable, useful quantum computing is achievable.

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